Github
Linkedin
LeetCode

Dragoljub Đurić

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Programming languages:
Proficient: C++ | Rust
Familiar: Go | Java | SQL

Software Engineer specializing in distributed systems, scalability, and performance optimization. Strong foundation in theoretical computer science and algorithm design, with experience in information retrieval, execution runtimes, and storage infrastructure.

Exp

Software Engineer — **Google - Core Search Infrastructure** — **Zürich** — **09/2025 – Present**

Retrieval Engine - the core of the search engine indexing trillions of documents with tens of millions QPS at ultra-low latency

- Slashed p99 latency to under 30s for the retrieval engine's update-validation tool across all search corpuses (including Google Search, YouTube, Gemini, and Cloud). Achieved this by fully distributing request workloads and eliminating OOM errors, rescuing a system that previously had a 90% failure rate due to 300s timeouts.

Software Engineer — **Dfinity** — **Zürich** — **10/2022 - 08/2025**

Internet Computer Execution Layer [dfinity/ic]

- Optimized **sandbox caching** by replacing LRU policy with a scheduler-aware eviction algorithm, aligning cache retention with execution priority. This stabilized the cache hit rate to **100%**, resolving a major performance bottleneck and contributing to a **4-5x** increase in system throughput [Middleware '25]
- Introduced the `cycles_burn128()` API, which has generated **95%** of the Internet Computer's execution **revenue** since launch
- Developed the **low_wasm_memory** hook to trigger actions when operative memory available for additional allocation drops below a threshold [demo]

Decentralized Storage Infrastructure – Founding engineer

- Architected a **distributed blob storage** system specifically optimized for the high-throughput, resilient data workloads of applications powered by LLMs and foundation models.

Stable Structures - Rust library of data structures operating in persistent memory [dfinity/stable-structures]

- Enhanced **BTreeMap** performance by **eliminating unnecessary value reads**, accelerating **20%** of operations and reducing execution time by up to **95%**
- Implemented and integrated **BTreeSet** data structure in the Stable Structures library, expanding its capabilities for efficient and reliable set operations in stable memory

Research assistant — **Oracle Labs** — **Zürich** — **07/2021 - 10/2021**

Oracle Cloud Infrastructure - Big Data Studio

- Developed a **performance analyzer** and sped up endpoint benchmarking by **250%** in Big Data Studio

Teaching assistant — **EPFL - École Polytechnique Fédérale de Lausanne** — **02/2021 - 07/2021**

- Led sessions for 200+ students on computability and computational complexity in **Theory of Computation** (CS-251)

Education

MSc in Computer Science — **EPFL - École Polytechnique Fédérale de Lausanne** — **09/2019 - 02/2022**

- GPA: **5.16/6** | Thesis: **Double Choco is NP-complete**
- Specialization: **Computer Science Theory**

BE in Information Systems — **University of Belgrade** — **10/2015 - 09/2019**

- GPA: **9.86/10** | Thesis: **A novel algorithm for sorting in linear time**
- Graduated as **Best in Class** out of 800+ students

Publications

- [Middleware '25] **Blockchain Inception: Scheduling and Caching Optimizations for supporting a PoW Blockchain on the Internet Computer**
  M Arutyunyan, A Berestovskyy, A Bratschi-Kaye, D Đurić, YA Pignolet, D Sarlis, A Sinpalean, A Uta
  ACM/IFIP Middleware Conference (Industry Track), 2025. doi:10.1145/3721463.3772059
- **Double Choco is NP-complete**
  D. Đurić
  arXiv preprint, 2022 arXiv:2203.02815
  Presented at the Theory Coffee Seminar, EPFL 2022

Awards

- **Dositeja Fellowship** — **EPFL and University of Belgrade, 2019–2022**
  - Serbia's highest national fellowship for academic merit, awarded for four consecutive years
- **ACM-ICPC Southwestern Europe Regional Contest (SWERC)** — **EPFL, 2019/20**
  - Ranked **31st** in the Final
- **INFORMS O.R. & Analytics Competition** — **General Motors, 2019**
  - **4th Place Finish (Finalist)** out of 290 teams for the "Redefining Vehicle Delivery with Autonomous Cars" challenge
  - Architected a multi-stage system using machine learning for demand forecasting to inform a route optimization algorithm
- **Dean's List** — **University of Belgrade, 2017–2019**
  - Awarded for academic excellence by placing in the top 1% of students